CHAPTER THREE

The Proteomics and Transcriptomics of Early Host Colonization and Overwintering Physiology in the Mountain Pine Beetle, *Dendroctonus ponderosae* Hopkins (Coleoptera: Curculionidae)

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Abstract

Two important phases in the life of the mountain pine beetle, *Dendroctonus ponderosae* Hopkins (Coleoptera: Curculionidae), are the periods of host colonization, where they must overcome host defences to successfully reproduce, and larval overwintering, during which time the insects under the bark of the tree must endure extended periods of deep cold. Proteomic and transcriptomic research into these portions of the mountain pine beetle life cycle have shed light on physiological processes that were already known to play important roles (detoxification of host secondary metabolites, reproductive physiology, and biosynthesis of cryoprotectants) and have revealed new territory for research (immune response, stress physiology). We discuss our overall findings in the context of an insect that is spreading to a new geographic range and encountering new hosts and a changing climate.

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1. INTRODUCTION

The mountain pine beetle is mainly a semelparous organism (Safranyik and Carroll, 2006), which, like many other organisms including some vertebrates such as salmon, reproduce once in their lifetime (although there are variations, eg, Reid, 1958 and references therein). After maturing during the course of a year under the bark of a host pine tree, new adult mountain pine beetles leave the relative protection of the host tree in which they matured and fly to find a new host in which to reproduce. As with salmon, this migration of sorts—more accurately a dispersal (Safranyik et al., 1992)—from a depleted natal host to a fresh, new host is fraught with danger. The insect does not feed during its dispersal flight (Reid, 1958). The adult life can be abruptly cut short due to the vicissitudes of weather: short rain squalls, summer wind storms, extreme heat, and unexpected cold. Predators, adapted to hone in on mountain pine beetle pheromone signals, stalk the boles of trees on which it lands. And in a mixed forest of nonhost conifers (Gray et al., 2015; Pureswaran and Borden, 2005; Pureswaran et al., 2004) and angiosperms (Huber and Borden, 2001, 2003; Huber et al., 2000), it needs to find one of the often few trees in the stand that will be suitable and susceptible to attack (Raffa and Berryman, 1983).

Mountain pine beetles, like many other bark beetles, have evolved powerful aggregation and antiaggregation pheromone systems (covered elsewhere in this volume) that guide many individuals to the correct hosts and that provide for coordination of mass attacks on host trees. This provides the insects with some level of advantage in the battle against the tree. In combination with detection of, and response to, volatile compounds emanating from host (Conn et al., 1983) and nonhost trees (Huber et al., 2000), as well as conspecific-produced pheromones, many mountain pine beetles often converge on one or a few trees in a stand (Hynum and Berryman, 1980; Lindgren and Borden, 1993; Pitman and Vité, 1969). Together, they stand a better chance of overcoming the tree's copious defences and successfully reproducing (Raffa and Berryman, 1983).

But even when the insect finds a host tree, the danger has not passed. A tree, like any other organism, has a vested interest in its survival. Because trees do not move after they germinate from a seed, they have necessarily evolved strong physical and chemical defences against all manner of enemies. And pines are well adapted to cope with bark beetle attacks, releasing both constitutive and induced resin-based defences against the invading insects and their often pathogenic symbionts (Franceschi et al, 2005; Raffa et al., 2005; Seybold et al., 2006; covered elsewhere in this volume). Tree defences are often classified as either 'constitutive' (Wittstock and Gershenzon, 2002) or 'induced' (Karban and Myers, 1989). Constitutive defences are those that are continually in place, as opposed to induced defences that are stimulated by insect attack or pathogen infection and are detectable in the plant only after such an assault. Many secondary metabolites that are present in the resin of a tree prior to attack by an insect are constitutive defences, as are physical barriers like the outer bark and other structural material that an invading insect must contend with. As mountain pine beetles attack, the tree releases resin stored in ducts (Franceschi et al, 2005; Reid and Watson, 1966). The resin is both a physical barrier to entry by the beetles and a mixture of chemicals, some of them toxic to the insect or its symbionts (Seybold et al., 2006). During this host colonization phase—a phase which we will examine in detail in terms of gene expression later in this chapter—invading mountain pine beetles need to cope with chemical defences, produce aggregation and antiaggregation pheromones, and complete their reproductive activity. Thus a mountain pine beetle, again somewhat like a salmon, enters an inhospitable environment and must survive long enough to reproduce while surrounded by competitors rushing to do the same thing. The process of dispersal and host colonization, while as short as mere hours or days at the most (Safranyik et al., 1992), is a crucial moment in the life of a mountain pine beetle. Insects that fail at this, either through maladaptation or simple bad luck, do not pass their genes along to the next generation.

If a pair of beetles is successful in host colonization, the female will lay eggs along the walls of a gallery constructed in the phloem (Cerezke, 1995). Larvae will soon hatch and will begin to feed on bark and phloem tissues that are replete with the secondary metabolites left behind from the tree's battle with the larvae's parents and their compatriots (Clark et al., 2012). While the larvae undoubtedly receive some nutrients from the maternal provisioning of the eggs, they need to ingest substantial amounts of toxin-laden phloem tissue in order to begin development and to get ready for the impending winter. The deep cold of winter, lasting up to a half year or more in some locations within the insects' geographical range, is often a major mortality factor for larvae (Cole, 1981). As with the dispersal and host colonization process, the ability, or lack thereof, of immature mountain pine beetles to survive the winter is both a major factor in natural selection (covered elsewhere in this volume) and in the growth dynamics of populations (Hicke et al., 2006;

Régnière and Bentz, 2007; Safranyik et al., 2010). A second part of this chapter will look in closer detail at larval overwintering physiology.

Because adult host colonization and larval overwintering are such vital points both for individual insects and in terms of the ecological and economic impact of expanding populations, recent research in our laboratory has targeted these processes. We have used new transcriptomics and proteomics methods to examine which genes are being expressed in insects during these times, and how that expression results in the accumulation of physiological relevant proteins.

Transcriptomic methods are powerful tools in advancing our understanding of organismal biology that allow us a window into the genes that are being expressed in an organism or tissue under the influence of various endogenous or exogenous factors. With these methods, we can assess an instant 'photograph' of expressed genes and compare that photograph to others at different times. However, just because we can determine which mRNAs are being produced in an organism or one of its tissues at a certain time, we do not know how (or even whether) those transcripts are being processed into their enzymes and other protein products. For the most part, it is the complement of proteins in a cell determines how that cell functions in the organism.

Parallel proteomic investigations, therefore, have the potential to help better interpret transcriptomic results. However, proteomics studies are also not without their limitations. For instance, a transcriptomic analysis may show increases in a particular mRNA, but a subsequent proteomic analysis on the same tissue may show no significant change in the level of the protein product. If a protein, vital to that organism at that time, is being rapidly produced and degraded while providing its metabolic function to the cells where it is expressed, then the proteomic result would not reflect the importance of that protein in the organism, but the transcriptomic result would. In addition, limitations of the sequencing methods and databases available for protein identifications from samples result in far fewer identified proteins in comparison to the gene transcript identifications. In each sample, we were able to annotate over 800 proteins based on the sequences obtained and on the available databases. In comparison, we were able to align the sequenced transcripts (many with an annotated gene function or name) of over 13,000 gene models from the mountain pine beetle genome Keeling et al., 2012, 2013a,b,c). Thus because of the sequencing technology as well as the available databases, our transcript data sets contain over an order of magnitude more information than we could glean from the proteomic data. But, by using both methods, in parallel when possible, we can better elucidate physiological function.

In recent work, we have used transcriptomic and proteomic methods on male and female adults attacking host lodgepole pine (compared with starved males and females), and larvae over the course of a winter (comparing larvae in early autumn with late autumn, and early spring with late spring). In each case, we used the same sample of insects for both proteomics and transcriptomics analyses; we were comparing apples to apples. So we feel that our four data sets represent a substantially complete picture of the physiological workings of these insects during these two life phases.

Specifically, we examined whole-body transcriptomes and proteomes in the two ecophysiological contexts. In the host colonization experiments, male and female mountain pine beetles that were collected in their natal bolts in southern British Columbia were emerged from their expended host material in our laboratory. Control males and females were then starved for 24 h prior to being extracted for transcriptomic and proteomic analysis. Treatment females were placed into predrilled holes in fresh lodgepole pine bolts and were partnered with a male. The pairs were then screened within the bolt and were removed 24 h later. Insects that showed evidence of feeding in the phloem were used in the transcriptomic and proteomic analyses. Statistical comparisons were made between starved and attacking individuals and between males and females. All treatments (male/female, starved/colonizing) were replicated four times.

In the larval overwintering work, we collected larvae from whole trees fitted with temperature dataloggers in the eastern central interior of British Columbia, near the Alberta border in the autumn of 2008 and spring of 2009. Some insects from the larger sample pool were used in specific transcriptomic work on the glycerol biosynthetic pathway (Fraser, 2011), and results from that work, along with knowledge of the temperature regime at the site that year, led us to choose four groups of insects from the sample pool for more extensive transcriptomic and proteomic studies. The larvae used in the study were collected in September and November (autumn) and March and May (spring). In general, our statistical analyses compared shifts in gene expression and protein levels across the autumn and across the spring. All treatments (dates) were replicated four times.

As with studies of this type that produce copious amounts of data (significant and otherwise), choosing which stories to tell and which to omit depends both on the patterns in the data and, to an extent, on the metaphorical lenses donned by the researchers. Others, doubtless, would look at the same data and see other complementary stories, and we certainly invite interested researchers to do that. In addition, we had to decide whether to extract RNA and proteins from whole insect bodies or to dissect out particular tissues for examination. While investigation of specific tissues would provide additional detail, we opted for whole-body extractions in order to gather a full, but perhaps less detailed, picture of physiological events in the insects. Given what we have found from this broader approach, we are now able to design more specific follow-up studies.

In general, we observed, and describe here, evidence of stress physiology, detoxification processes, pheromone biosynthesis, immune function, and rapid shifts in reproductive physiology during host colonization; along with detoxification, stress physiology, cold hardening, and developmental processes during larval overwintering. Over the course of this chapter, we will detail some of our findings that are particularly relevant to these general categories in the two life phases. In all cases, unless otherwise noted, where we refer to genes or gene products expressed in our experiments, we are citing Bonnett et al. (2012) (overwintering proteomics), Robert et al. (2013) (host colonization transcriptomics), Pitt et al. (2014) (host colonization proteomics), and J.A. Robert et al. unpublished data (overwintering transcriptomics).

2. HOST COLONIZATION

Trees maintain a variety of formidable defences and vigorously resist attacks by bark beetles and their associated microorganisms (Franceschi et al, 2005; Raffa et al., 2005; Seybold et al., 2006). The best-studied chemical defences, in terms of their effects on bark beetles and their symbiotic fungi, are the terpenoids (for review, see Bohlmann, 2012). The defensive aspects of conifer terpenoids are made more complex by the fact that bark beetles often detect them alone or in conjunction with aggregation pheromone components emitted by other colonizing conspecifics to locate host trees (Seybold et al., 2006). In addition, many bark beetle pheromone components are similar to host terpenoids.

Upon encountering a host tree—found through a combination of visual cues (Campbell and Borden, 2006; Shepherd, 1966), avoidance of non-hosts (Huber and Borden, 2001, 2003; Huber et al., 2000; Pureswaran and Borden, 2005; Pureswaran et al., 2004), detection of host volatiles (Billings et al., 1976; Wood, 1982), detection of con- or heterospecific pheromone emissions (Byers, 1989), and direct testing of candidate trees (Raffa and Berryman, 1982a,b)—an attacking bark beetle first encounters the outer bark of the tree. Bark beetles maintain a number of wood-degrading enzymes

(Pauchet et al., 2010) and we detected some of these present in the bodies of colonizing insects. We found that some are upregulated during the early attack phase. Plant cell wall–degrading enzymes are a group of enzymes defined by their ability to break down the polysaccharide (eg, cellulose) and glycoprotein components of plant cell walls (Pauchet et al., 2010). These components are part of the physical defences of the host tree and one of the barriers that mountain pine beetle must contend with in order to successfully reproduce in a selected host tree.

In our study of the physiological changes in adult mountain pine beetle during early host colonization, we saw the upregulation of transcripts coding for a variety of plant cell wall degrading enzymes (eg, glucanases, endopolygalacturonases, pectin methylesterases), and this upregulation is different between males and females. Females are the first to colonize and chew into a suitable host tree, and it is in the females that we see the largest change and variety in the expression of plant cell wall degrading enzyme transcripts, several of them increasing over 50-fold above the levels measured in the control females. In our proteomics analysis, both males and females had detectable plant cell wall degrading enzymes, but the relative values of the proteins did not show the dynamic range that was apparent in the transcript expression data. The presence of these enzymes during early host colonization indicates some level of constitutive expression. In other words, bark beetles are primed to begin to digest tree tissue prior to their arrival at the tree. And indeed, our control beetles (ie, those not yet exposed to host plant tissue) show detectable expression of transcripts annotated as plant cell wall degrading enzymes. This is consistent with the attacking bark beetles needing to feed in their natal host and then bore through the outer bark to emerge for dispersal. The same enzymes that would aid in those predispersal tasks would be used during host colonization.

In the mountain pine beetle, the female is the pioneering sex (Latty and Reid, 2009) and is the first to arrive at a host tree to begin the process of excavating through the outer bark and phloem. As the first arrivers, they are more susceptible to harm from early constitutive defences (Latty and Reid, 2009). In addition to producing the cell wall degrading enzymes that allow them to overcome the host tree's physical barriers to attack, they also produce the initial components of the overall pheromone bouquet that will emanate from a tree under attack (Libbey et al., 1985; Pureswaran et al., 2000). Males and other females detect these emissions within the mix of the host volatiles released by female excavation and within the bark and phloem. Upon joining the aggregation, they release pheromone components, with the male bouquet differing from that of the female in composition and timing (Pureswaran et al., 2000). At some point in the host colonization process, when the insects detect that the tree has lost the ability to effectively defend itself, or when an optimum number of conspecifics are present in the aggregation, an antiaggregation pheromone—in this case verbenone—is produced (Hunt and Borden, 1990; Hunt et al., 1989; Pitman et al., 1969; Rudinsky et al., 1974; Ryker and Yandell, 1983). The dynamics of the physiological and biosynthetic processes of pheromone biosynthesis by bark beetles is outlined elsewhere in this volume. In our studies, transcript changes reflecting the shift in pheromone production within the first 24 h of beetle colonization was primarily evident in male beetles. We observed a male-specific increase in the expression of transcripts encoding two keys enzymes in the mevalonate pathway—3-hydroxy-3-methylglutaryl-CoA synthase and 3-hydroxy-3-methylglutaryl-CoA reductase. This suggests a physiological shift from the production of *exo*-brevicomin (Song et al., 2014a,b), a pheromone that attracts females (Conn et al., 1983; Rudinsky et al., 1974), to frontalin, an antiaggregation pheromone produced from isoprenyl precursors made by the mevalonate pathway (Keeling et al., 2013a,b,c). Thus a significant rise in the key enzymes in the mevalonate pathway supports the shift in male pheromone production during the first 24 h of host colonization from the production of chemicals to attract females to one that deters competition.

It should be noted that our work only covered the very first hours of host colonization, and that in a somewhat contrived situation in which females were forced into a particular location on the bark in cut bolts rather than standing trees, and where neither sex was able to exercise any mate choice (Pureswaran and Borden, 2005). In addition, all of the insects that we assayed for comparison against controls were in the presence of the opposite sex. Thus further detailed work could be done on long-term trends in gene regulation during the full course of a typical colonization event—for example, late colonization and the release of antiaggregation pheromone components—ideally in a more natural setting, along with gene regulation of the two sexes not in association with each other. As detailed in other chapters in this volume and elsewhere (eg, Keeling et al., 2013a,b,c; Song et al., 2014a,b), a great deal of this information is already known for genes specifically related to the main pheromone biosynthetic pathways, thanks to much previous and ongoing excellent work in this field, but there doubtless still many aspects that remain to be uncovered in this complex and dynamic phenomenon.

Upon breaching the barrier of the outer bark, females, and later the males who join them, encounter phloem saturated with secondary metabolites (Seybold et al., 2006). Phloem tissue contains resin ducts (Franceschi et al., 2005; Reid and Watson, 1966), some of which the insects may sever, often resulting in insects doused with resin and literally swimming in the sticky and toxic fluid (Keeling et al., 2006; Raffa and Berryman, 1982a,b). Again, because our experiments were limited to the first day of host colonization and because we were using freshly cut bolts (and hence likely only constitutive defences were present as cut trees could not respond to attacks), the effects that we saw should be seen as limited to the insects dealing with constitutive chemical defences. Many of the same enzymes may be operative in detoxification of both constitutive and induced secondary metabolites. First, while there are qualitative differences between plant constitutive and induced defences, including pine (Ott et al., 2011), many of the differences are qualitative in nature. Thus the insects deal with many of the same secondary metabolites—albeit in differing amounts—in the early constitutive phase as in the later induced phase. Second, some detoxification enzymes are known to function on a variety of related substrates (Feyereisen, 1999). A classic example of this is some of the cytochromes P450 in parsnip webworms that have been shown to each detoxify several of the furanocoumarins in their wild parsnip hosts (Li et al., 2004). Other examples of this sort exist—including in taxa as disparate from mountain pine beetles as humans (Danielson, 2002)—so it is likely that at least a subset of the proteins and unregulated genes that we detected in the early colonization phase are also part of the detoxification machinery in later colonization. One antennal-specific cytochrome P450 is known to use pine terpenoids as a substrate (Keeling et al., 2013a,b,c), so such activity is not unexpected elsewhere in the insect's body.

Mountain pine beetle adults do not often have second chances at mating once they have entered a host tree. There are likely exceptions to this, for instance D.H. and others have observed an early emergence of previous year adults that, particularly in large infestations, have the potential to colonize and reproduce in a second tree (and see Reid, 1958 and associated references). In addition, in other bark beetle species, such as *Ips pini* (Reid and Roitberg, 1995), males are known to abandon galleries after mating to search for more mating opportunities in the same year. In any case, it is likely that host colonizing mountain pine beetles are investing as much of their energy as possible in their reproductive opportunity and therefore any energy invested in the metabolism of toxins reduces reproductive output. Thus there is likely a balancing act between detoxification and excretion of byproducts, and overall tolerance of the host defence compounds.

Evidence for the tolerance of tree-produced toxins may be seen in the proliferation of a variety of detoxification and stress physiology-related gene regulation and protein products. In our work, 24 h after exposure to host tissue, the transcripts for cytochromes P450, transferases, esterases, ABC transporters, and alcohol dehydrogenases showed large and significant changes. Each of these could play a role in the detoxification of host tree defence compounds and enable the attacking beetles to complete their reproductive cycle. Cytochromes P450 are very common enzymes that are often implicated in the detoxification of a wide range of substrates in insects (Feyereisen, 1999; Li et al., 2004); several have been identified that play a role in the detoxification of terpenoids and in pheromone production in bark beetles (Aw et al., 2010; Keeling et al., 2006; Tittiger et al., 2005). In our study of mountain pine beetle adults during early host colonization, the gene transcripts encoding for cytochromes P450 showed a significant and varied change for both male and female beetles. Cytochromes P450 are notoriously difficult to assign a function based on sequence information (Feyereisen, 1999), but it is likely that several of the cytochromes P450 that were observed to increase following feeding, especially those associated with the midgut and fat body, play a role in the detoxification of host defence compounds. Similarly, glutathione-*S*-transferases and glucosyl/glucuronosyl transferases alter the solubility of substrates to facilitate excretion from the insect body, and they have been shown to function in concert with cytochromes P450 to detoxify xenobiotics (Sheenan et al., 2001). Both male and female beetles showed significantly increased expression of several transferases within 24 h of exposure to host tissue. An identified glutathione-*S*-transferase protein was also elevated in males relative to controls in our proteome data. In addition, transcripts annotated as esterases—enzymes implicated in insect resistance to insecticides (Li et al., 2007)—showed increased expression in female beetles. An ABC transporter, similar to those generally known for their role in sequestration of toxic compounds (Labbé et al., 2010), also increased significantly in females after exposure to host tissue in our experiments. Because a portion of toxic pine resin is comprised of terpenoid alcohols, the observed upregulation of transcripts annotated as alcohol dehydrogenases in both males and females may also play a role in aiding the adult beetle to survive long enough to mate, excavate galleries, and lay eggs.

The presence of stress physiology-related proteins and upregulation of such genes indicates the possibility of some secondary metabolites affecting vital biochemical processes by direct effect on proteins. While the specific effects of diverse conifer secondary metabolites on protein or other targets in bark beetles requires more investigation, we do know that some plant secondary metabolites specifically target protein function (Acamovic and Brooker, 2005). Some of the stress physiology-related proteins—heat shock proteins (Feder and Hofmann, 1999) and calreticulin (Fink, 1999), for example—that were strongly upregulated shortly after exposure to the host are known to specifically act to maintain correct protein folding. Other upregulated genes and protein products are known to be involved in protection against cellular oxidative damage—eg, superoxide dismutase (Lopez-Martinez et al., 2008)—again pointing to potential effects of secondary metabolites on portions of the cellular machinery.

It is possible that an upregulated stress response is present in colonizing bark beetles, not only in the context of stress related to exposure to host toxins but also to help them to deal with the effects of substantial flight exercise during dispersal. Our experiments did not subject the insects to precolonization flight, so if this is the case, then perhaps some stress-related proteins are upregulated immediately following flight due to exogenous or endogenous signals related to the initiation of host colonization. That is, a stress response related to investment in repair and maintenance of tissues following the dispersal flight might only be initiated upon the completion of successful host finding when bark beetle flight muscles are known to be catabolized for energy (Bhakthan et al., 1970; Reid, 1958; Sahota, 1975).

In any case, future work targeting the effects specific resin chemicals that are likely to interact with proteins would contribute to helping us understand tree defence and would even potentially reveal new and useful pharmaceuticals for human use. The stress response in host-attacking mountain pine beetles is, to our knowledge, not discussed elsewhere in the literature and certainly deserves further work that will lead to a better understanding of host colonization success and failure in this and related species.

While the presence of an upregulated stress response in host colonizing adults was somewhat surprising to us, even more surprising was the existence of what seems to be a substantial immune response. For example, transcripts annotated as C-type lectins and scavenger proteins were upregulated in both male and female beetles. These groups of enzymes are responsible for the identification of foreign cells slated for destruction by the immune system (Tanji et al., 2006). Transcripts annotated as having a role in the identification of bacterial invaders or in an antiviral response showed significant upregulation. Almost no information exists for the role of these transcripts in insects, and so the impact of the beetle's immune response during host colonization is an area that is currently wide open for study.

Just as a variety of predators and parasitoids feed on bark beetles, it is also highly likely that pathogens, spread prior to emergence from the host tree or encountered on the bark of potential hosts or during mating encounters, opportunistically or specifically infect adult bark beetles. Thus in addition to the stress of dealing with a dispersal flight and host tree defences, colonizing bark beetles could also face pathogenic agents that could reduce their lifespan and fecundity in what is likely their sole mating attempt. In addition, depending on the pathogen, an infected adult may spread disease to offspring that either affect their growth and development or that persists with them until adulthood to infect the next generation.

Immunity in bark beetles remains understudied, or more to the point, mainly unstudied in the fullest sense. Bark beetles, including mountain pine beetles, do interact with a variety of microorganisms. Of particular note are the symbiotic and pathogenic fungi (Reid et al., 1967; Whitney, 1971) that are often carried in specialized structures called mycangia. In an early large-scale proteomic study of southern pine beetle (*Dendroctonus frontalis*) mycangia proteome, Pechanova et al. (2008) found a large number of proteins with a variety of annotated functions associated with those structures. One overall message from that study was that the insects are not merely receptacles and vehicles for fungal spores but also likely play some as yet to be determined role in the life cycle and physiology of the microorganisms (Adams et al., 2013) that they carry. That is, while their symbiotic fungi doubtless affect bark beetle physiology and behaviour, it seems that the tight connection between the insect and the fungi is a two-way street. Immunological functions could play a role in this partnership by, for instance, regulating the phenology and activity of associated symbiotic or other microorganisms that are known to impact reproductive success of adults or their larvae as well (Therrien et al., 2015).

The role of bacteria and viruses in the reproductive success of mountain pine beetle is particularly interesting considering recent discoveries about vitellogenin (Salmela et al., 2015). In our work, we noted that vitellogenin was upregulated by 1400-fold (transcriptome) and 2.3-fold (proteome) in female mountain pine beetles almost immediately upon encountering host material. Work by Salmela et al. (2015) in honeybees has shown that vitellogenin binds to pathogens and to other metabolic indicators of pathogen presence. Upon being deposited into eggs, the vitellogenin carries parts of these pathogens with them, and thus has the potential to be the source for intergenerational immune priming in insects. If this is the case in bark beetles as in honeybees, then the sudden onset of vitellogenin synthesis in females only after the dispersal flight makes sense in an immunological context as that is when the female is most likely to be encountering pathogens that may be detrimental to her young.

Having found a host tree, entered it, encountered the tree defences and associated pathogens, deposited and perhaps helped to activate symbiotic fungi, mountain pine beetles begin to excavate an ovipositional gallery and lay eggs. There has been a substantial amount of research on adult condition as it relates to flight exercise and later reproductive output. Our results, particularly the massive upregulation of vitellogenin and its precursors, indicate that females do not commit energy to producing reproductive products (ie, provisioned eggs) until they are sure that they do not require further energy for dispersal. Other evidence that points toward this temporal partitioning of resource use come from past work on degradation of wing muscles following initiation of host colonization in some bark beetles (Reid, 1958). We observed a differential production of protein products involved in cytoskeletal and anatomical maintenance and rearrangement such as an increase in calponin (an enzyme that plays a role in the regulation of muscle contraction) with a simultaneous decrease in actinin and kinesin (proteins that play a role in the structure of muscle tissue) in adult males after exposure to host tree tissue.

Having now successfully deposited fertilized eggs, a pair of bark beetles has completed that portion of their life cycle. However unless their soon-to-hatch larvae also survive to make the same difficult journey, the work will have been in vain.

3. LARVAL OVERWINTERING

Mountain pine beetle eggs are deposited in small niches on the sides of the ovipositional gallery and hatch shortly thereafter. The very young larvae begin to mine through the phloem tissue, moving perpendicularly from the vertical ovipositional gallery and feeding as they go. While much attention has been paid to the tree defences encountered by the attacking parents, work in our research program has recently shown that the larvae encounter higher levels of some secondary metabolites than their parents (Clark et al., 2012). This makes sense in the context of constitutive and induced defensive responses by attacked hosts. When the parental generation first encounters the tree, they encounter only the constitutive defences.

The tree, detecting the attack, begins to synthesize and excrete induced resin and, in the case in other conifers (Nagy et al., 2000), modify its anatomy to more effectively permeate its affected tissues with secondary metabolites (Reid and Watson, 1966; Reid et al., 1967). This response is rapid enough that even though in a large scale mass attack the tree will still die, but prior to death, the phloem becomes highly saturated with resin compounds. These compounds are variably volatile and do not necessarily quickly dissipate from the tree. Some of fungi that were inoculated into the tree by the attacking adults are known to be able to detoxify some conifer secondary metabolites (Hesse-Orce et al., 2010), but growth of the fungi takes some time. Thus, larvae are born into a potentially toxic wasteland that has been left behind after the chemical warfare conducted by the host tree against their parents.

This presents a substantial problem for the larvae. While they hatch in the heat of summer, winter is always fast approaching; summer can cool to autumn temperatures prior to the equinox, and as the days shorten, a sudden cold snap could arrive at any time. The early instar larvae need to develop overwintering defences against early cold snaps and the later deep and extended cold of winter in just a few short weeks of feeding in host phloem, and they need to do this in an environment and food source that is saturated with toxins left over from their parents' battle with the tree.

It is therefore likely that detoxification of host secondary metabolites is a vital process in the larvae as well as in the adults. While the adults must cope with constitutive defences and the early onslaught of induced defences, once they have successfully deposited eggs in the gallery, their task is completed. And while some adults survive winter and may reemerge early the following year (D. Huber, personal observation) or even in the same year (Reid, 1958), most do not. As such, we expected to find substantial evidence of detoxification metabolism in larvae.

In the autumn larvae, no less than 11 transcripts annotated as cytochromes P450 increase significantly in expression between September and November. All of these show no change over the winter, with the exception of two that show a threefold decrease between November and March, and all show either no change or a significant decrease in spring. In addition, two glutathione-*S*-transferases, enzymes involved in the transfer of a glutathione moiety to potentially toxic chemicals entering the insect body in order to increase hydrophilicity and excretion, changed significantly over the winter in both the gene expression and proteomics study. In the transcriptomics study, two putative glutathione-*S*-transferases increased significantly in expression between September and November, and the expression of both genes decreased in the period between March and May (but the decrease is only significant for one). This significant decrease was echoed in the proteomics data for a detected glutathione-*S*-transferase in the spring.

In general, we observed maintenance of constitutive levels of proteins but increased expression of genes involved in detoxification activity during the course of the autumn—cytochromes P450, glutathione-*S*-transferases, and esterases. In the spring, we observed—with a few exceptions—decreases in the levels of protein representatives of these gene families across the course of the season. This indicates that by the time that we assayed larvae (late-September) they had already built up their defences against host toxins. This is logical, of course, because by this time, the larvae would have been feeding in the phloem tissue for several weeks or even several months. Not having the ability to feed and remove toxins would have proven fatal. On the other hand, the maintenance or drop in detoxification-related expression and gene products in the spring coincides with the likely drop in the levels of phloem-based secondary metabolites due to natural chemical and physical processes and the detoxification activity of beetle-associated symbiotic fungi. If, in fact, levels of secondary metabolites remaining in host tissue are lower in the spring (something that we have yet to test), then the maintenance of high levels of detoxification enzymes would represent a metabolic expense at a time when completing development in a timely manner for subsequent dispersal and reproduction is important.

Much like the adults, the larvae seem to be coping with and preparing for physiological stress during the autumn. Many of the same stress physiology-related genes and protein products that we saw expressed in host colonizing adults, plus some others, were either constitutively expressed in autumn larvae or upregulated in autumn larvae and downregulated in spring larvae. These include heat shock proteins, proteins associated with stressors such as heat, cold, and desiccation as they help stabilize proteins within the cell (Denlinger and Lee, 2010). Expression of two transcripts annotated as heat shock proteins shows a pattern of increase in autumn, stability over the winter, and then a significant decrease in spring. In addition, the largest fold change in expression that we observed for larvae between September and November was again annotated as a galactose-specific C-type lectin, an enzyme associated with immune response in insects (Tanji et al., 2006). This gene transcript increased over 100-fold in the autumn. This suggests that, not only are the developing larvae exposed to the threat of toxic host compounds and abiotic stressors but also to microbial pathogens.

Constitutively expressed stress physiology genes are likely to be, at least in part, responsible for helping the larvae to cope with resin toxins as it feeds. The various proteins and genes that showed increased levels of expression between September and November are more likely to be involved in dealing with oxidative stress and the presence of free radicals brought about by cold temperatures and subsequent physiological processes (eg, dehydration) that might be involved in the cold-hardening response (Denlinger and Lee, 2010). At the very least, we can say that the stress response in autumn mountain pine beetle larvae is real and substantial, although at this point, we do not know the specifics of its function. The drop in stress-related gene expression and protein levels in the spring, like the drop in detoxification metabolism, also makes sense if levels of secondary metabolites and substantial risk of deep cold have dissipated at this time. In terms of an energy budget, spring is the time for the larvae to spend their energy and any metabolic surplus left over from the winter on growth and development in order to be ready for their dispersal flight and subsequent chance at reproduction in the coming summer.

An obvious physiological shift that should be required for autumn mountain pine beetle larvae is the development of cold hardening. Mountain pine beetles are not freeze tolerant; that is, they cannot allow their bodily fluids to freeze. Instead, as with other freeze resistant insects, they employ several strategies, including the accumulation of antifreeze compounds in their haemolymph to reduce their supercooling point (Bentz and Mullins, 1999). The supercooling point is the point, below the normal freezing temperature of water, where the haemolymph will freeze, and mountain pine beetles are known to be able to reduce this to around −35 °C (Safranyik, 1998), and perhaps colder in some cases.

One of the main cryoprotectant compounds that has been observed in overwintering mountain pine beetles is glycerol (Bentz and Mullins, 1999), and our results showed an increase in the level of glycerol biosynthetic proteins (and related gene expression) in the autumn as well as a subsequent drop in the same proteins and expression in the spring. In one of the pathways leading to the production of glycerol, the transcripts for two key enzymes in the catalytic steps immediately preceding the synthesis of glycerol (glycerol-3-phosphate dehydrogenase and glycerol kinase) were significantly upregulated in autumn larvae between September and November. These same enzymes show a large drop (decreasing 10- and 6-fold, respectively) between samples of larvae taken in March and May the following spring.

These results closely match those of other work in our laboratory (Fraser, 2011) in which real time quantitative PCR was employed to monitor gene expression specific to most of the steps in glycerol biosynthesis from glycolysis onwards. The insects used in the overwintering proteomic and transcriptomic studies were from the same population (geographically and temporally) as those used in the RT-qPCR work. Those insects, collected near to Valemount, British Columbia, also represent a seeming transitional population in terms of their population genetics (Gayathri Samarasekera et al., 2012), although this was not known at the time of collection. Thus while the glycerol results coincide well with our expectations (ie, increases and decreases of cold-hardening gene expression in autumn and spring, respectively), it is possible that this trend is not going to be identical for all populations across the much larger range of this insect. While we chose our samples for analysis based on the temperature regime measured at the trees—using samples taken prior to and after the first substantial cold—it is not yet known if exogenous signals (temperature, light levels filtering through the bark, etc.) or endogenous signals (developmental, circadian/circannual physiology, etc.) or a combination are responsible for the timing of these processes.

Other cryoprotectant metabolism also showed similar autumn/spring shifts in gene expression. For instance protein levels of 2-deoxyglucose-6-phosphate phosphatase, an enzyme that produces 2-deoxyglucose, increased substantially in the autumn and decreased very dramatically in the spring. 2-Deoxyglucose is known to regulate glycolysis and cell growth (Barban and Schulze, 1961; Wan et al., 2004)—upstream reactions from glycerol biosynthesis—and could itself also be a cryoprotectant. α,α,-Trehalose-phosphate synthase also showed significant increases in protein level in the autumn and significant decreases in the spring. While trehalose generally makes up a large portion of the haemolymph carbohydrates in insects, it is also a cryoprotectant (Baust and Lee, 1983). From these results it is likely that glycerol is not the only cryoprotectant employed by mountain pine beetle larvae during autumn cold hardening.

Larvae that are able to survive and adequately feed in toxin-laden phloem in the autumn, that endure the stress of those toxins and of overwintering, and that produce enough cryoprotectants to avoid freezing in the depths of the winter must then continue their growth and development in anticipation of summer dispersal and reproduction. Gene expression shifts noticeably from detoxification, immune response, and stress response during the autumn to growth and development in during the spring thaw.

Of the transcripts that show greater than 10-fold decrease in expression in the autumn, eight are annotated as insect cuticle proteins. Conversely, in the spring, one of the transcripts showing over 1000-fold increase is annotated as a digestive enzyme in beetles, cathepsin L-like proteinase (Murdock et al., 1987). We found several dozen growth and development-related proteins in autumn larvae, but only a caprin homologue (cytoplasmic activation/proliferation-associated protein) showed a shift in levels between September and November (a small but significant decrease). We also found several dozen (a few more than in the spring) proteins in the same category in the spring, and in this case, a dozen of them showed significant increases between March and May and two others showed substantial and significant decreases during the same time period. In the latter case, the two enzymes had annotations similar to juvenile hormone epoxide hydrolase, indicating a likely shift in juvenile hormone titres during later development. Many of the transcripts that show large and significant increases in the spring are unannotated—a great deal of work needs to be done in order to understand the physiological processes occurring in mountain pine beetle as they come out of overwintering and begin the process of pupation and emergence.

A good summary of mountain pine beetle overwintering might be to say that autumn is a time of intense and frantic preparation for winter done within the still-hostile environment of the dead host. In the spring, the surviving larvae must complete development with access to what appears to be a more benign food source, and they reallocate their metabolic expenses away from stress physiology, detoxification, and cold hardening and towards development. In essence, just as with adult host colonization, this phase is a dynamic and active one for the insect in terms of physiology, and we have a great deal more to learn about the numerous processes that have been partially uncovered by this recent research.

4. IMPLICATIONS

Due to the economic and ecological impacts of mountain pine beetle activity, the volume of past research on this species has possibly made it one of the better known insects in the world. Following the massive outbreaks in central and northern British Columbia and parts of the United States during the turn of the current century, the concentration of available research funding, and thus effort, has revealed more than we could have imagined in a variety of biological (and beyond) contexts. The research and ideas outlined in this chapter, elsewhere in this book, and throughout the recent literature, provide massive amounts of information for guiding policy and management, as well as many new questions to be answered. Substantial funding and subsequent research effort by many excellent scientists was no accident—it was in large part due to the anticipation of movement of the beetle from its historical redoubt west of the Rocky Mountains in British Columbia into the boreal forests of Alberta.

Around 2006, the first established populations of mountain pine beetles were observed in lodgepole pine forests in the northeast corner of British Columbia, a region that is on the eastern slope of the Rocky Mountains and that had not previously been exposed to the insect (Janes et al., 2014). During the next several years the beetle moved through the eastern population of lodgepole pine, into a region in west-central Alberta where lodgepole and jack pine substantially hybridize, and from there into pure jack pine stands in central and east-central Alberta (Cullingham et al., 2011). In addition to this movement of the outbreak into a new host and a new geographical region to the east, the insect is also currently expanding its range further north in British Columbia and Alberta towards to pine forests of Yukon Territory and the Northwest Territories.

These range shifts are unprecedented in recorded history and we have no reason to believe that they do not represent a completely unique phenomenon, at least since the previous ice age. To put it succinctly, we seem to be witnessing the movement of an important North American species into new hosts and into a new range where it will doubtless experience a novel climate. Mountain pine beetle is a native species to this continent, but it also represents an invasive species on a (large) regional scale. A number of factors likely at play in this movement include climate change affecting both temperature regime and tree vitality and defence; human activity such as the suppression of fire and creation of large even-aged stands through logging practices; and year-by-year meteorological conditions during the flight period. These factors, and others, are reviewed and assessed elsewhere (eg, Anderegg et al., 2015; Janes et al., 2014; Keane et al., 2015; Sambaraju et al., 2012; Tobin et al., 2014).

The question then becomes, 'what can be done about this?' The most likely answer is, frankly, 'not much' if what is meant is direct control and eradication of the insect in its new range. If climate is the major driver of the phenomenon, then without immediate and radical shifts in fossil fuel use, the factors that may reduce or stop the spread of this insect in its new range are novel tree defences in the new host, stand structure as it affects dispersal in the new geographic range, or ideally timed and severe cold snaps over large land areas. Even with radical curtailment of carbon and other greenhouse gas emissions, a great deal of the future warming and other climatic change, particularly at the higher latitudes at which this portion of the mountain pine beetle infestation is active, is already inevitable.

There are three foreseeable and logical outcomes to the current spread of mountain pine beetle into jack pine forests, none of which are under direct human control. First, perhaps there will be a substantial cold snap in the new range in a near future winter while the beetle's foray into its new range is still not too extensive. Perhaps it will be deep cold in the mid-winter, or substantial cold in the mid-autumn or early spring when the larvae have reduced levels of resistance to cold or stress. The progression of climate change, however, likely means fewer deep cold events in North America (Screen et al., 2015), so the odds of climate-based solutions worsen each year, even while the infestation in the new range potentially covers more territory.

Second, perhaps the new jack pine host maintains defences that slow beetle population growth substantially, or perhaps the nutrients in jack pines are insufficient for growing larvae to produce appropriate overwintering defences. In terms of host defence in jack pine at this time, from what we know (Clark et al., 2014; Lusebrink et al, 2011; Taft et al., 2015), the trees are likely at best equally, and at worst, less defended from mountain pine beetles than are lodgepole pines. Beyond that, mountain pine beetles are able, in some circumstances, to successfully aggregate, mate, and reproduce even in some nonpines (Huber et al., 2009) demonstrating a substantial level of flexibility in host use. Almost every pine species in their large geographical range—other than a few exceptions—is susceptible to them (Furniss and Carolin, 1977; Gray et al., 2015). There is no reason to expect that jack pine, which easily hybridizes with lodgepole pine, will not be a usable host. We do still need to learn more about the mountain pine beetle in its new range, but at this point, it is reasonable to assume that there is potential for an outbreak in jack pine stands similar to the one that occurred in lodgepole pine in British Columbia and elsewhere.

Third, mountain pine beetle in the jack pine boreal forest may be the new normal. If that is the case, then we will have to learn to manage it there, as silviculturists have done in lodgepole pine forests for generations. Development of models (Shore and Safranyik, 1992) of susceptibility, risk, and spread based on mountain pine beetle physiology as it relates to cold tolerance and host colonization will be vital to managers in this new region (Aukema et al., 2008; Safranyik et al., 2010). Our proteomics and transcriptomics results provide new hints of the important aspects of mountain pine beetle physiology to monitor and study on a deeper level, and give a research into management and the development of new models a further leg-up as the insect and humans both move into uncharted territory.

5. CONCLUSION

Our first proteomic- and transcriptomic-based look at mountain pine beetle physiology, following the development of the genomic data required for its analysis, has shown that adult insects colonizing trees put a premium on detoxification, resistance to stress, immune function, and conservation of as much energy as possible for reproductive output. Larval insects feed in highly toxic tissues and seasonally adjust their stress physiology, detoxification machinery, and developmental processes in response to host toxins and the onset and departure of winter cold. During both life phases and stages we observe very dynamic gene expression and, presumably, concomitant physiological shifts. Our work has provided a large number of target genes and proteins that are now available for study related to mountain pine beetle physiology in the context of the identity, quantity, and timing of defences in a new host; climate in a new geographical range (depth of cold, timing of cold, interaction of winter onset, and host quality); and the potential encounter of new pathogens, or new interactions, that might further affect immune and stress responses.

We look forward to further work with some of these processes in our research program, and to hearing about work by others, as we move forward in understanding and managing this important, yet also very interesting, insect.

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